Tutorial - 11

7/4/26

1 Dataset

Sample	Feature 1 (Age)	Feature 2 (Income)	Feature 3 (credit Score)	Actual	Predicted Prob
S_1	25	40	650	1	0.90
S_2	30	60	700	1	0.75
S_3	45	35	600	0	0.60
S_4	50	30	580	0	0.40
S_5	28	55	720	1	0.30

1 → positive
0 → negative.

If probability ≥ 0.5 - 1
" < 0.5 - 0

Actual (y)	predict (based on 0.5)
1	1 (TP)
1	1 (TP)
0	1 (FP)
0	0 (TN)
1	0 (FN)

1. Use threshold Condition 0.5 → confusion matrix
2. Calculate F1 score, Recall, precision, Accuracy
3. Calculate False positive rate (FPR)
4. Compute the TPR & FPR for threshold 0.7, 0.6, 0.5, 0.4 & 0.3
5. Plot the Roc points / curve

Confusion matrix

Predicted \ Actual	1	0
1	TP 2	FP 1
0	FN 1	TN 1

→ Accuracy

$$\frac{Tp + TN}{\text{total no.}} = \frac{2+1}{5}$$

$$= 3/5$$

$$= 0.6$$

⇒ precision

$$\frac{Tp}{Tp+Fp} = \frac{2}{2+1} = 2/3 = 0.6$$

⇒ Recall (TPR)

$$\frac{Tp}{Tp+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.6$$

⇒ F1 Score

$$\frac{2 \times \text{Recall} \times \text{precision}}{\text{Recall} + \text{precision}} = \frac{2 \times 0.6 \times 0.6}{0.6 + 0.6}$$

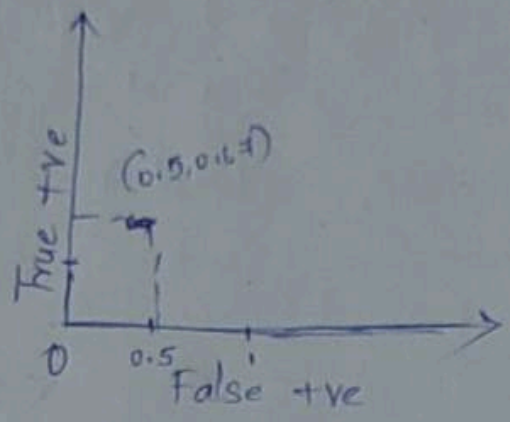
$$= \frac{0.8}{1.2} = \frac{\cancel{8}\,2}{\cancel{12}\,3} = 0.6$$

⇒ False positive (FPR)

$$\frac{Fp}{Fp+TN} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

⇒ True (TPR) $= \frac{Tp}{Tp+FN} = 0.67$

→ Receiver operating characteristics (ROC)



→ Take threshold probability 0.3

Actual	Probability	
1	1	(TP)
1	1	(TP)
0	1	(FP)
0	1	(FP)
1	1	(TP)

Predicted ~~Actual~~	1	0
1	TP 3	FP 2
0	TN 0	FN 0

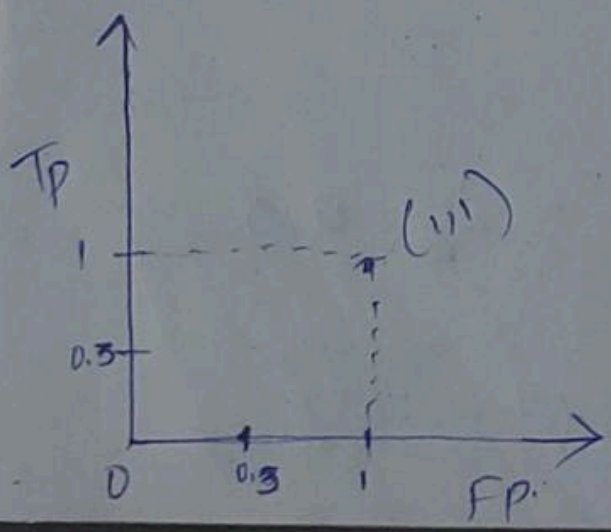
~~Predicted~~ Actual

→ FPR

$$= \frac{2}{2+0} = 1$$

⇒ TPR

$$= \frac{3}{3+0} = 1$$



Threshold 0.4

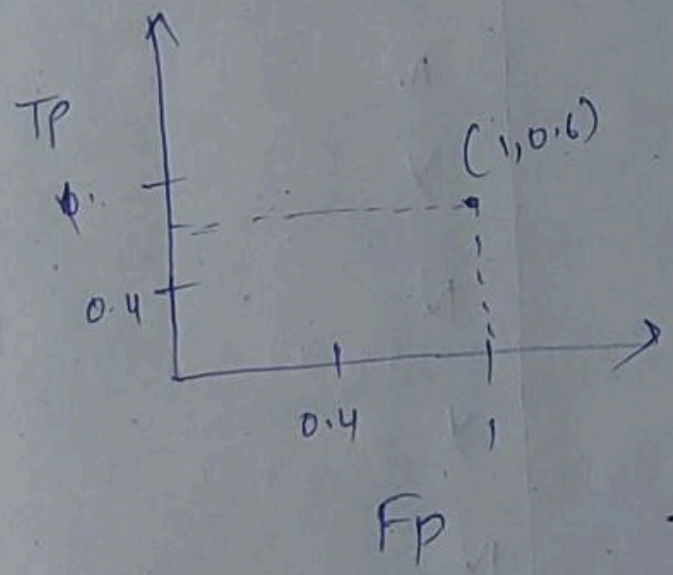
Actual	Predict	
1	1	(TP)
1	1	(TP)
0	1	(FP)
0	1	(FP)
1	0	(FN)

Confusion matrix

Predicted \ Actual	1	0
1	TP 2	FP 2
0	FN 0	TN 1

→ FPR

$$= \frac{2}{2+0} = 1$$

→ TPR

$$= \frac{2}{2+1} = \frac{2}{3} = 0.6$$